

(U) Request for Information (RFI)
Regarding
Post Quantum Encryption in Microelectronics

(U) This Request for Information (RFI) is in accordance with Federal Acquisition Regulation (FAR) clause 52.215-3 (Request for Information or Solicitation for Planning Purposes - Oct 1997) and is published to obtain information for use by the Air Force Lifecycle Management Center (AFLCMC), Cryptologic and Cyber Systems Division (HNC), Joint Base San Antonio – Lackland AFB, Texas, for market research and planning purposes only.

(U) This RFI is NOT a Request for Proposal, Invitation for Bid, or an announcement of a solicitation; it is only intended for information or planning purposes. There is no bid package or solicitation document associated with this announcement. Response to this RFI is strictly voluntary and will not affect any potential offeror's ability to submit an offer if a solicitation is released. Any requests for a solicitation package will be disregarded. The government does not intend to award a contract on the basis of this RFI or otherwise pay for the information solicited. No entitlement to payment of direct or indirect costs or charges by the government will arise as a result of preparing submissions in response to this RFI and the government's use of such information. Submittals will not be returned to the sender. Respondents to this RFI may be requested to provide additional information/details based on their initial submittals.

Program

(U) This RFI is in support of programs within the Cryptologic Modernization Branch (AFLCMC/HNCA).

Purpose

(U) This RFI is NOT a Request for Proposal, Invitation for Bid, or an announcement of a solicitation; it is only intended to seek information from industry. The information provided may be used by the program office to develop its acquisition strategy, statement of work and/or system requirements. Respondents to this RFI may be requested to provide additional information/details based on their initial submittals.

Background

(U) The rapid advancement of quantum computing poses a significant threat to the cryptographic algorithms that underpin modern cybersecurity, prompting a growing interest in post-quantum algorithms designed to withstand attacks from quantum-capable adversaries. As quantum computers become more powerful, traditional encryption methods, such as those based on elliptic curve cryptography, risk being rendered obsolete due to their vulnerability to quantum algorithms like Shor's algorithm. In response to this emerging challenge, the National Institute of Standards and Technology (NIST) has taken proactive steps to develop and standardize post-quantum cryptographic (PQC) algorithms; NIST has recently released the standards for three new cryptographic algorithms (FIPS-203, -204, and -205) designed to be secure against quantum computing threats.

(U) The United States Air Force (USAF) develops cryptographic devices which are certified according to the National Security Agency's (NSA) high-assurance certification process. This

process ensures that cryptographic devices used to secure classified communications and data will not be vulnerable to adversary threats. Certification of a high-assurance cryptographic device entails evaluation of all physical and logical security mechanisms for protecting sensitive data and the processing of that data, to include cryptographic algorithms implemented in hardware and software. Designers and manufacturers of cryptographic devices implement these functions in microelectronics within the device, known as the “crypto module” or “crypto engine.” The crypto module may be an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or a combination of microprocessors known as a “system-on-chip” (SoC). The crypto module contains root-level firmware which typically includes some cryptographic functions to verify the authenticity of root-level processes such as secure boot. These cryptographic functions, at the crypto module level, will be the focus of this RFI.

Information Sought

(U) AFLCMC/HNCA requests industry provide information on the items listed in the table below.

Request ID	Request
R001	Describe your role in the production of cryptographic devices (e.g. developer, system integrator, microelectronics vendor, microelectronics manufacturer).
R002	Provide the name, version, and release year of all cryptographic modules or engines you manufacture.
R003	Specify whether each cryptographic module or engine is ASIC-based or FPGA-based.
R004	List any microelectronic families that your company manufactures that perform cryptographic functions within your crypto modules.
R005	List any microelectronic families that perform cryptographic functions (e.g. FPGAs) that are sourced from a third-party. Indicate whether these microelectronic families are commercial or government-off-the-shelf, or custom developed.
R006	Identify the manufacturer of each chip used for cryptographic processing.
R007	Describe any cryptographic functionality (e.g. signatures, boot verification) included in FPGAs by the microelectronics manufacturer that are used in your crypto modules. Does your designs leverage any built-in cryptographic functionality provided by the embedded FPGAs (e.g. signatures, boot verification)?
R008	Describe any additional security features at the root-level of the microelectronics within your cryptographic modules. Indicate whether “out-of-the-box” security features are reprogrammed before/during embedment in a crypto module.

R009	Describe any modifications made to commercial microelectronics used in your crypto modules to support quantum resilience.
R010	Does any of your devices currently support hybrid cryptographic modes combining classical and PQC algorithms? If yes, describe the implementation.
R011	Provide the timeline for integration or public availability of PQC capabilities in your products.
R012	Do you allow customers to program custom cryptographic functions onto your platforms? If yes, describe the process and limitations.
R013	Describe the biggest challenges in adopting PQC algorithms at the device/hardware/firmware level.

(U) Vendors who submit information for review do so with the understanding that the U.S. Government personnel as well as their support contractors (MITRE, will review their material and data. Respondents will not be contacted regarding their submission or information gathered as a result of this notice nor the outcome of the Government's review of the solicited information unless the Government desires further information. Submitted information packages will be retained by the Government and not returned to the contractor. Ensure information is marked and sent appropriately if the response is For Official Use Only. If your response is classified, contact the point of contact listed below before forwarding the information.

(U) All interested, capable and responsible sources that wish to respond to this RFI are required to electronically supply their responses to Ms. Erin Azar @ erin.azar@us.af.mil by COB 29 April 2025. Please ensure responses are 8.5x11, single-sided pages and include the title, "*Post-Quantum Encryption in Microelectronics RFI Response*," in the Subject Line. Telephonic responses will not be accepted.

(U) Please include with your submittal, a cover page that includes your Company Name, Commercial and Government Entity (Cage) Code, Mailing Address, Country of Ownership, and the point of contact for Cryptographic Development.

NOTE: Additional follow-up questions may need to be accomplished over a secure form of communications.